

CDU IT CODE FAIR 2026: DATA INNOVATION CHALLENGE

The Long Way to Care

Mapping the Mobile Coverage and Emergency Access Gap Across the Northern Territory's Remote Communities

A Composite Risk Index Linking Phone Coverage, Clinic and Hospital Access, and Road Isolation for 782 Remote Communities

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Summary

A mobile phone signal can decide whether someone in remote Northern Territory community is able to call for help and access emergency care. The Long Way to Care project built a transparent Gap Index that score 782 remote communities on four factors: mobile coverage, drive time to the nearest clinic, drive time to the nearest hospital, and access to a sealed road. The project combined nine government, telecommunications, and health sources. Of the 782 communities, 429 (54.9%) have no indicated means of making an emergency call. Risk is concentrated: 196 communities sit in the Critical tier, and 98% of them are family outstations, small kin-based homelands, not towns. The result is stable, because alternative weightings and a settlement-type variant identify the same top 20 communities (Spearman $\rho \geq 0.99$). Ranking clinics by the risk in their catchments, instead of the number of communities served, lifts Amoonguna, Minjilang, and Ramingining into the top ten. The burden falls mainly on Aboriginal communities living on their homelands, so the analysis uses no personal data, scores missing information as a risk rather than discarding it, and should be verified with communities before it guides decisions. We recommend risk-weighted clinic outreach, priority blackspot investment for Critical-tier outstations, contingency communication plans for known blackspots, and community-led verification of coverage data. An offline interactive dashboard puts the results in the hands of health services and communities.

1. Introduction

The most important thing to do in any emergency is to call for help. For some part of the Northern Territory, we can't just assume that that's an option. Long distances and low population in our cyclone- and flood-prone Territory, combined with patchy mobile coverage, leave many remote communities unable to call for help by phone, let alone reach a clinic or hospital far more quickly once that help is on its way.

The Long Way to Care was built to make this problem measurable and visible. Instead of treating phone coverage, clinic distance, hospital distance, and road access as separate pieces of information that cannot be assessed together, we combine them into a single score: the Gap Index. A transparent, reproducible way of rating just how likely it is for any singular community to reach for help in an emergency successfully. It's not an attempt to diagnose why coverage may be so poor, but to provide health planners, telecommunications providers, organizations, and the government with evidence-based knowledge necessary to decide which communities should take priority.

The scope of this analysis comprises 782 NT communities (ranging from remote outstations to regional towns) and 91 remote and regional health clinics. We used current 2025 mobile coverage data alongside government registers of territories and communities, clinic and hospital location data from the Northern Territory Government, road network data, cyclone exposure history, and socioeconomic details. This is not just a clinical risk model; it is specifically a screening and prioritization model - identifying only where combined barriers are the worst so that the limited investment in towers, staffing, or preparedness would be channeled to where it is most needed.

2. Methodology

2.1 Data sources and integration

Nine sources were cleaned and combined into two tables: a community table and a clinic table. These included NT Government remote communities and 2022 mobile coverage listings from the NT Government, along with 2025 mobile network site registers for Telstra, Optus, Optus-TPG regional co-build (MOCN), and TPG, sourced from the ACCC's Mobile Infrastructure Report release. Additionally, the Geoscience Australia facility, the national road network data, and the Bureau of Meteorology provided cyclone exposure history. Further context was added using AIHW's potentially preventable hospitalization data and the Australian Digital Inclusion Index to provide socio-economic insights. Road network data was used to determine drive times, while location and name matching were used for community and clinic records. The clinics' phone number, hours, and 24/7 status were obtained from the NT Government remote health services list. To ensure consistent re-execution across different machines, all scripts in the processing pipeline use portable relative file paths.

2.2 Building the Gap Index

For each community, four risk factors were calculated and scaled from 0 to 1. A score of 0 means the community is in the best position, while 1 means it is at the highest risk.

The four risk factors are:

- **risk_coverage** – whether the community has a reliable way to call for help.
- **risk_clinic_time** – the driving time needed to reach a clinic.
- **risk_hospital_time** – the driving time needed to reach a hospital.
- **risk_no_sealed_road** – whether the community has access to a sealed road.

Where driving-time information was missing, it was mainly because calculating routes to very remote communities was difficult; the missing values were assigned the highest risk score of **1.0** rather than removed. This helps ensure that remote communities, which are important to this analysis, remain included.

The main **Gap Index** was calculated by assigning equal weight to all four risk factors and averaging them (equal-weighted mean).

An optional fifth factor, called **Index B**, was also tested. This used the rate of people in each settlement type who reported they **could not make a phone call (cannot call)**. The aim was to see whether the type of settlement provides additional useful information beyond the four main risk factors.

2.3 Statistical approach and validity checks

A Shapiro–Wilk test was conducted for each continuous variable used in the index (Table 1). All six variables deviated significantly from a normal distribution, with $p < 0.001$ for each. Therefore, non-parametric statistical methods were used throughout the analysis. Medians and interquartile ranges (IQRs) were used to summarise the data instead of means and standard deviations. Spearman's rank correlation was used to examine relationships among index components and to identify potential redundancy. For group comparisons, the Kruskal–Wallis test and Mann–Whitney U test were used instead of ANOVA and t-tests, respectively.

Variable	N	Mean	Median	Std Dev	Min	Max	Shapiro–Wilk p	Normal?
Drive time to nearest clinic (min)	763	69.50	45.00	81.25	0.00	581.00	3.4e-31	No
Drive time to nearest hospital (min)	697	252.58	236.00	167.30	1.00	751.00	3.3e-11	No
Distance to nearest clinic (km)	782	35.62	27.21	37.74	0.00	244.84	8.1e-30	No
Distance to road network (km)	782	1.06	0.10	4.37	0.00	69.20	1.8e-48	No
Cyclone count within 200km	782	16.42	20.00	12.96	0.00	44.00	1.3e-23	No
Severe cyclone count within 100km	782	0.67	0.00	1.12	0.00	4.00	7.3e-38	No

Table 1. Descriptive statistics for the six continuous variables feeding the Gap Index (n = number of communities with a recorded value; Shapiro–Wilk $p < 0.05$ rejects normality for all six).

Before developing the index, the relationships between the different measures of access and isolation were examined (Figure 1). The results showed a strong correlation between drive time to the nearest clinic and the straight-line distance to the clinic ($\rho = 0.88$). In comparison, clinic drive time had a moderate correlation with hospital drive time ($\rho = 0.44$) and a weak correlation with distance to the nearest sealed road ($\rho = 0.23$).

These results suggest that access to clinics and hospitals, and road isolation, are related but represent different aspects of community access and should not be treated as the same barrier.

A similar pattern was observed among the four core Gap Index components: **risk_coverage**, **risk_clinic_time**, **risk_hospital_time**, and **risk_no_sealed_road**. Their pairwise Spearman correlations ranged from **0.25 to 0.69**, indicating positive relationships but not strong enough to suggest substantial overlap. This supports the inclusion of all four components, as each provides distinct information about the different access and isolation challenges experienced by communities.

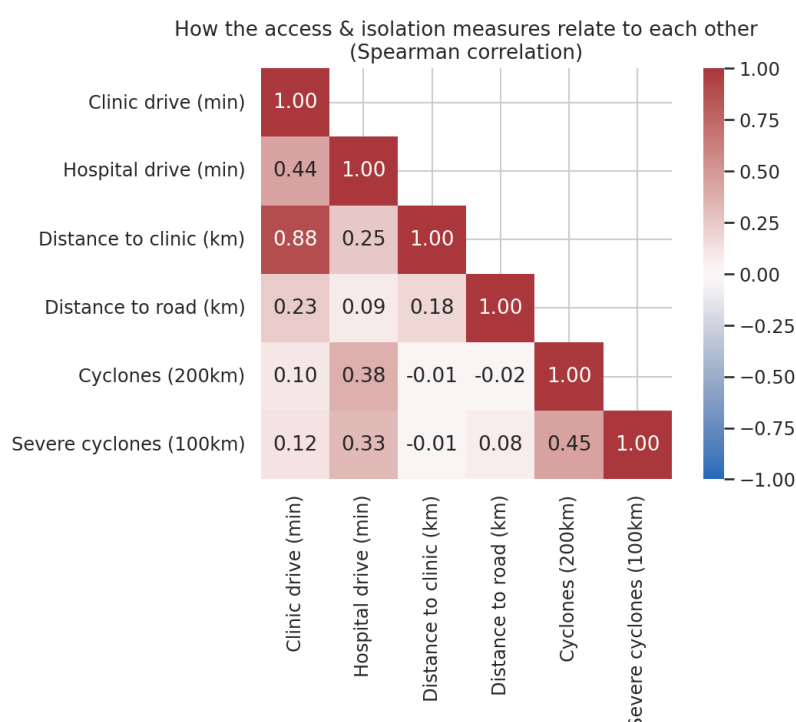


Figure 1. Spearman correlation between the six continuous variables used to build the Gap Index and describe community risk.

Two robustness checks were conducted before selecting the final index. First, a weight-sensitivity analysis compared the equal-weighted index with two alternative weighting approaches: one that gave greater weight to the drive-time components and another that gave greater weight to the coverage component. All three approaches produced the same 20 highest-risk communities, with very strong Spearman correlations ranging from $p = 0.991$ to 1.000 . This indicates that the results were highly consistent regardless of the weighting approach used.

Second, **Index A**, which included the four core components, was compared with **Index B**, which added settlement type as a fifth component. The two indices showed an almost perfect correlation ($p = 0.990$) and identified the same 20 highest-risk communities. This suggests that the settlement type provided very little additional information after the four core components were included.

Based on these findings, **Index A was selected as the final Gap Index** because it provides a simpler, more transparent measure and yields consistent results. Communities were then classified into four risk tiers—**Low, Moderate, High, and Critical**—using the quartiles of the Gap Index. This resulted in approximately **195–196 communities per tier**, as expected under the quartile-based classification.

3. Findings

3.1 The scale of the coverage gap

Of the **782 communities** analyzed, **429 (54.9%)** were in coverage categories with no confirmed way to make a phone call. This included communities where coverage information was not recorded in the register and those explicitly recorded as having no coverage. A further **177 communities** relied on

mobile towers recorded only in the 2025 carrier registers, while **161 communities** were located near a macro cell but did not have confirmed direct coverage (Figure 2). Overall, these findings indicate that for more than half of the communities analyzed, access to a phone call for emergency assistance cannot be assumed.

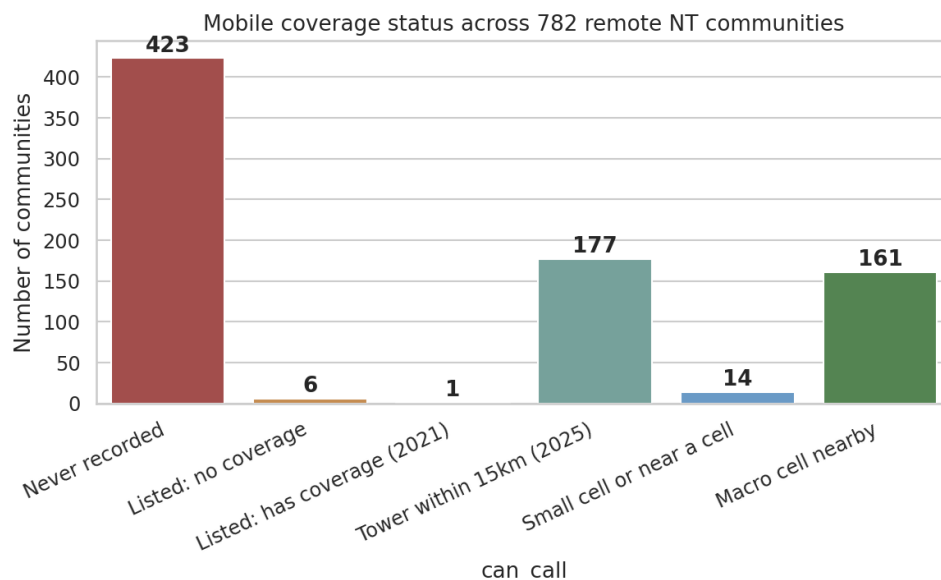


Figure 2. Mobile phone coverage status across all 782 NT communities analysed.

The Gap also varied across regions (Figure 3). East Arnhem had the highest average Gap Index, with a mean score of 0.720. Barkly had the highest proportion of communities with no confirmed way to make a phone call (65.2%), compared with 54.9% across the Territory. Big Rivers ranked closely behind on both measures. A Kruskal–Wallis test showed statistically significant differences in the Gap Index across coverage categories ($H = 438.54$, $p < 0.001$) and settlement types ($H = 290.06$, $p < 0.001$). Similarly, the Mann–Whitney U test revealed a significant difference in the Gap Index between communities with and without a confirmed way to make phone calls ($U = 130,447$, $p < 0.001$). These results indicate that coverage status is strongly associated with the overall Gap Index.

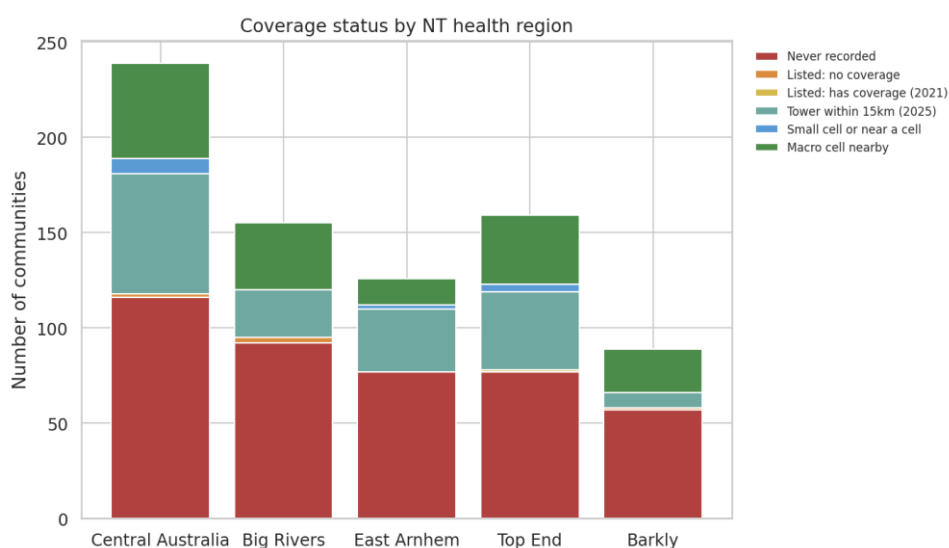


Figure 3. Mobile coverage category by NT health region. Central Australia has the most communities overall; Barkly and Big Rivers carry the largest share in the 'never recorded' category.

3.2 Who carries the highest risk

Table 2 lists 10 of the 16 communities that shares the maximum Gap Index score of 1.000. Figure A9 in Appendix C charts the highest-risk communities.

Community	Community type	Region	Gap Index	Risk tier
Kumunu	Family Outstation	Barkly	1.000	Critical
Natjitnama	Family Outstation	Central Australia	1.000	Critical
Nguyarramini	Family Outstation	Barkly	1.000	Critical
Mallarrami	Family Outstation	East Arnhem	1.000	Critical
Kudantiga	Family Outstation	Top End	1.000	Critical
Ngarinthe	Family Outstation	Top End	1.000	Critical
Namerinni	Family Outstation	Barkly	1.000	Critical
Ji-Malawa	Family Outstation	East Arnhem	1.000	Critical
Ankweyeyelengkwe	Family Outstation	—	1.000	Critical
Mu-Gurta	Family Outstation	Top End	1.000	Critical

Table 2. 10 of the 16 communities recording the maximum Gap Index score (1.000), all in the Critical risk tier. A dash means no region is recorded in the source data.

Kumunu, a Family Outstation in Barkly, illustrates what a maximum Gap Index score represents. It had no confirmed way to make a phone call, the longest clinic and hospital drive times in the dataset, and no sealed road access. In comparison, **Tennant Creek**, a town in the same region, had a Gap Index score of only **0.012**. Coverage was confirmed; its clinic drive time was longer than that of only 5% of communities; its hospital drive time was shorter than that of all other communities; and it had access to a sealed road. The difference between a Gap Index score of **0.012** and **1.000** within the same region demonstrates the substantial variation in access between communities across the Territory.

3.3 Rethinking clinic priorities

The 91 clinics were initially ranked according to the number of communities they serve. While this is a common way to identify clinics for service planning, it does not account for the different levels of risk faced by those communities. We therefore also ranked the clinics using the total Gap Index score across the communities in each clinic's catchment area. The two approaches produced broadly similar results, with seven of the ten highest-ranked clinics appearing in both lists. However, **Amoonguna**, **Minjilang**, and **Ramingining** appeared in the top ten when Gap Index risk was taken into account, even though they did not appear in the top ten based on the number of communities served (Figure 4). The two rankings had a strong correlation (**Spearman ρ = 0.916**). This suggests that the number of communities served does not always reflect the level of access difficulty within a clinic's catchment.

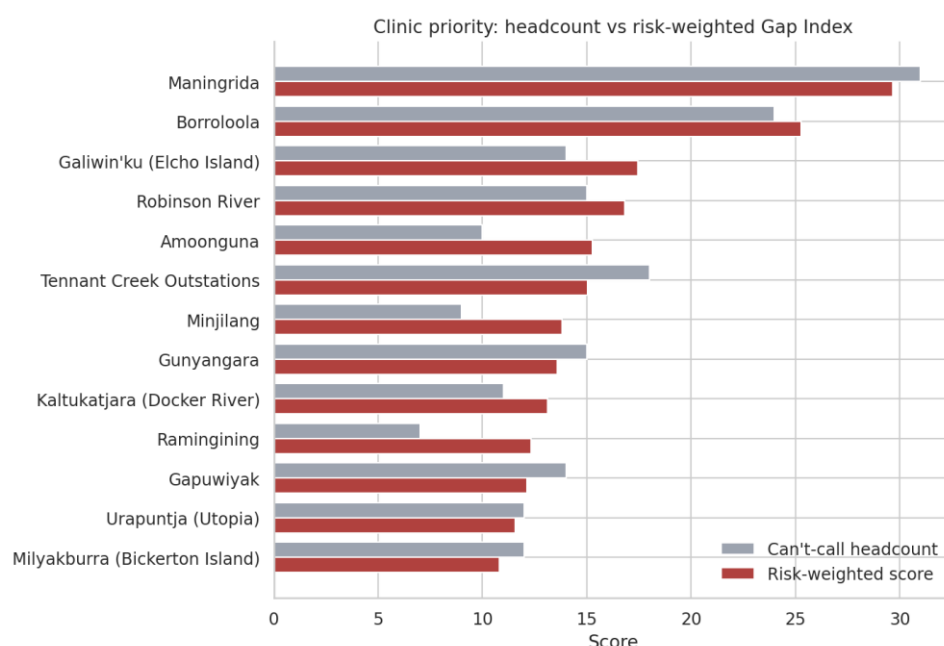


Figure 4. Clinic prioritisation by patient headcount versus total risk-weighted Gap Index. Amoonguna, Minjilang and Ramingining rank higher under risk-weighting than under headcount alone.

4. Discussion

The highest Gap Index scores were mainly found among Family Outstations. These communities are generally small and spread across remote areas, with many families living on their traditional country. Their geographic location and small population size can make it more difficult and costly to provide reliable telecommunications, roads, and health services. The findings, therefore, highlight the importance of considering how services can be improved for people living in these communities, rather than assuming that moving to larger centers is the only way to address access difficulties.

The findings should also be considered in light of several limitations in the data. The coverage information was obtained from government and telecommunications carrier registers. These records indicate where coverage is reported to be available, but they do not necessarily show the level of mobile signal experienced by people on the ground. Actual coverage can be affected by factors such as terrain, vegetation, and buildings. Another consideration is the treatment of missing drive-time values. Where a drive time could not be calculated, a maximum risk score of **1.0** was assigned. This was done because the missing routes were mainly associated with remote communities where road access was difficult to determine. Although this approach allowed these communities to remain in the analysis, the assigned value is a modeling assumption rather than a directly observed measurement.

The Gap Index also does not account for local arrangements that may already help communities respond to emergencies. For example, some communities may use **UHF radios, satellite phones, or community-based support arrangements** where mobile coverage is limited. These existing resources may reduce the risk experienced in practice, even when the Gap Index indicates a high level of access risk.

Ethical, cultural, and community considerations

The analysis uses only place-level public data. No personal or health records were used, and the clinic contact details are the published service numbers. The unit of analysis is a community's access to help, not a person, and the results describe gaps in infrastructure and services, not failings of the communities themselves. Many of the highest-risk places are homelands where families live on Country by choice, so the findings should not be read as an argument for relocation. Missing data are scored as maximum risk rather than dropped, and a place with no recorded coverage is described as 'never recorded', which does not imply that no signal exists. Because the data describe communities and were not collected with them, the results should be verified with community members, land councils, and Aboriginal community-controlled health services before they guide funding, in line with Indigenous data sovereignty principles such as the CARE principles. The dashboard runs on a device with no signal, so the people most affected do not need to be online to use it.

Overall, the **Gap Index is intended to support screening and prioritization rather than provide a definitive measure of clinical or individual risk**. It can help identify communities and clinic catchments where access barriers may be greater, but the results should be considered alongside local knowledge and community experiences. This would provide a more complete basis for decisions about future services and infrastructure.

5. Recommendations

For NT Health and remote clinic services

- When working with regions on prioritizing outreach visits and staffing, look beyond patient headcount at catchment risk-weighting to reach places like Amoonguna, Minjilang, and Ramingining that are priorities for risk weighting but not for patient headcount.

For government and telecommunications providers

- Prioritize Critical-tier Family Outstations when directing the Mobile Black Spot Program and equivalent infrastructure investment. East Arnhem, Barkly, and the Big Rivers stand out as the regions with the largest and longest lists of critical-tier outstations, according to our analysis.

For emergency services and community organisations

- Do not assume a mobile call is possible in an emergency; pre-register known blackspot communities (the 429 communities with no confirmed way to call) for satellite phones, UHF radio, or community-warden contingency protocols.

For future data and community engagement

- **Verify reported mobile coverage against community experience before making funding decisions.** Carrier and government coverage records should be supplemented with information from residents and local service providers. Any resulting changes to telecommunications or other services should also be planned in partnership with the communities affected.

- Plan to re-run the gap analysis annually as new carrier site registers and rollouts become available and extend it with clinical outcome data (such as potentially preventable AIHW hospitalizations) as it becomes available at community-level datasets.
- Distribute the offline The Long Way to Care dashboard to carriers, clinics, and communities so they can look up coverage and record coverage gaps directly, rather than relying solely on periodic government and carrier registers.

References

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Appendix A: AI Usage Declaration

Generative AI (Claude, Anthropic) was used as a copilot and learning support tool throughout the project. It assisted the team in interpreting project requirements, understanding analytical questions, developing and debugging Python code, and structuring the dashboard and report. All AI-assisted outputs were critically reviewed, tested, and validated by the team, who retained responsibility for the analysis, interpretation, discussion and final conclusions.

Appendix B: Source Code and Dataset Links

Source code (Python pipeline: data cleaning, merging, exploratory analysis, Gap Index model, dashboard build): <https://github.com/Cynthia-Ike/IT-Code-Fair-2026>

Interactive dashboard (The Long Way to Care), the offline file dashboard_offline.html in the Outputs folder: <https://github.com/Cynthia-Ike/IT-Code-Fair-2026>

Raw and processed datasets (Raw Data, Clean Data, Processed Data and Scored Data folders): <https://github.com/Cynthia-Ike/IT-Code-Fair-2026>

Appendix C: Supplementary Figures

The figures below support the findings in the main report but were placed here to keep the main body within the page limit.

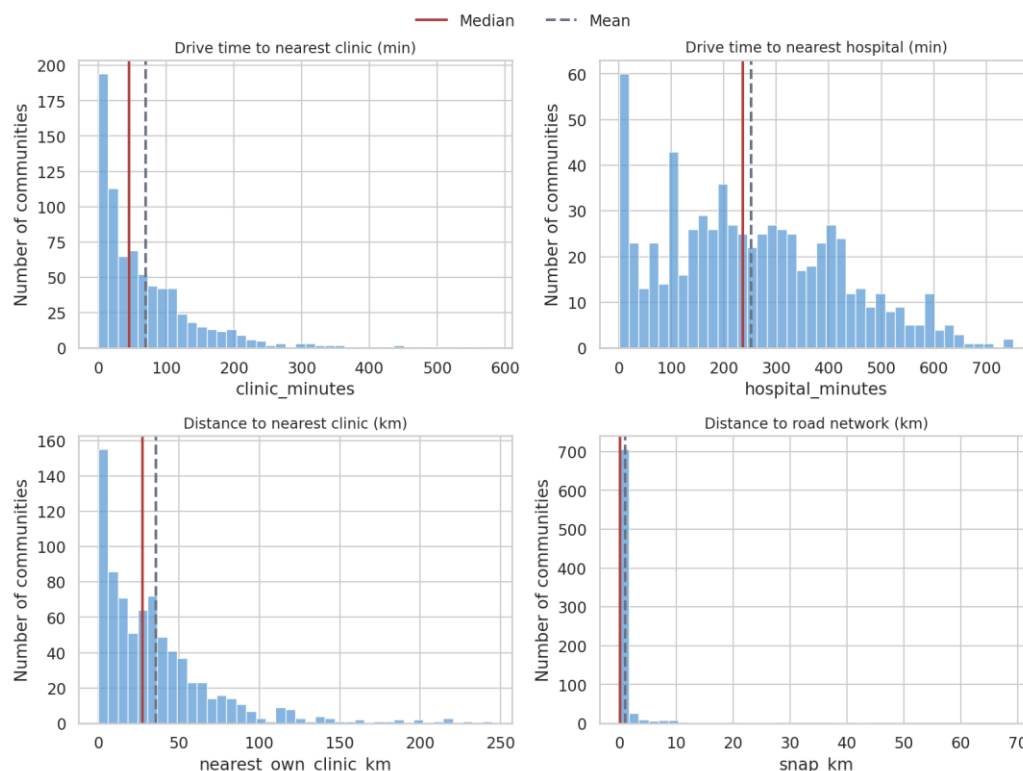


Figure A1. Distribution and normality check (Shapiro–Wilk) for the six continuous variables used in the Gap Index.

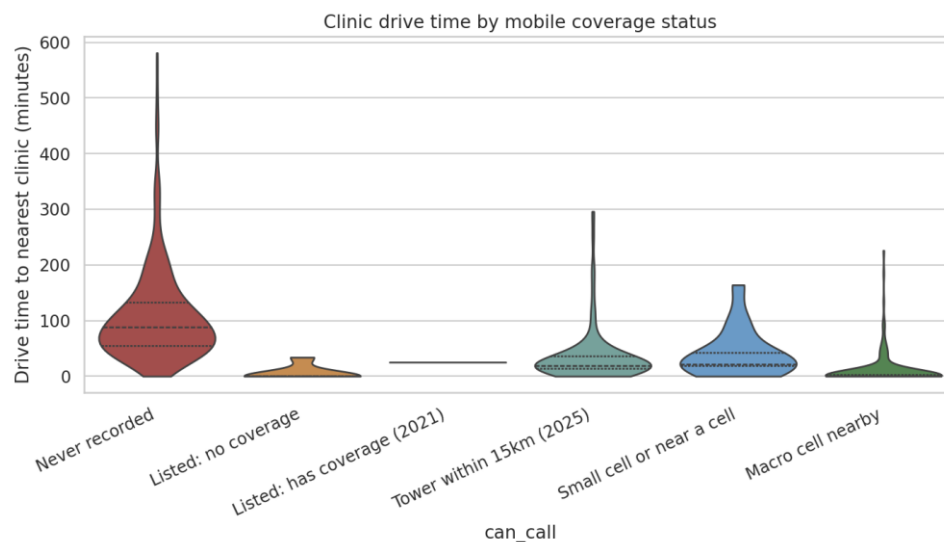


Figure A2. Drive time to nearest clinic, by phone coverage category.

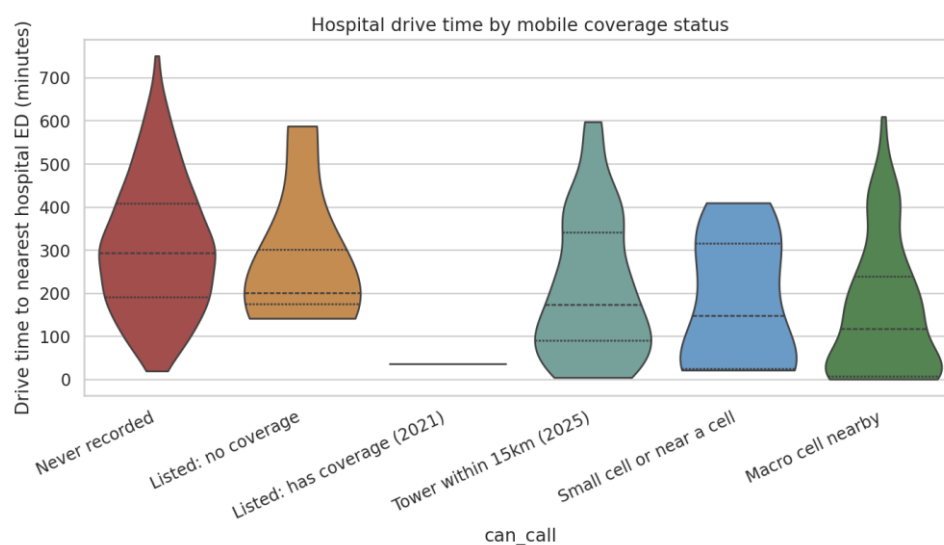


Figure A3. Drive time to nearest hospital, by phone coverage category.

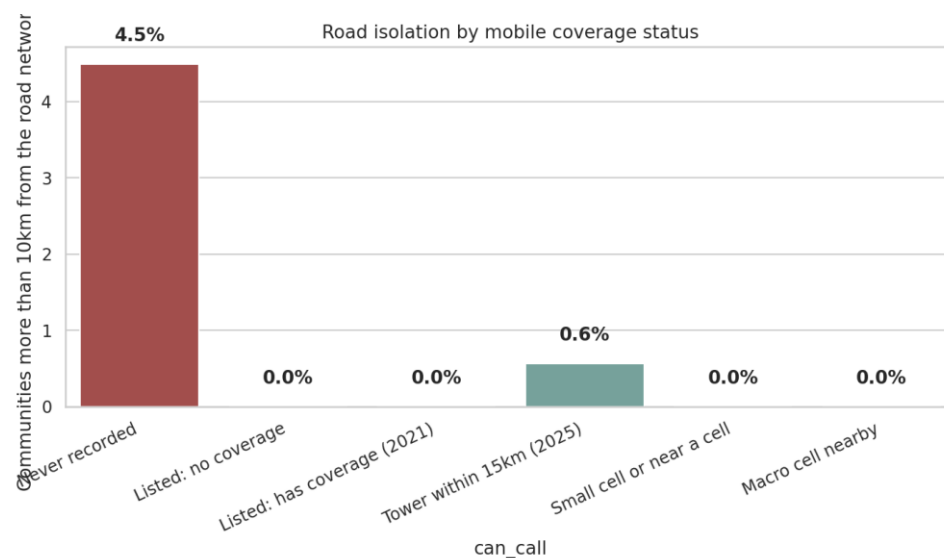


Figure A4. Road isolation (distance to nearest sealed road), by phone coverage category.

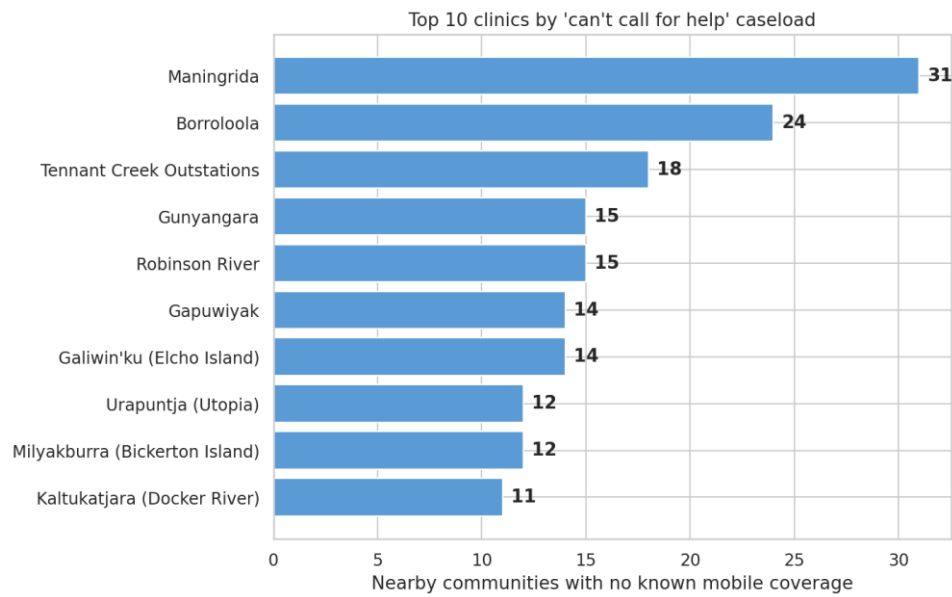


Figure A5. Community caseload per clinic.

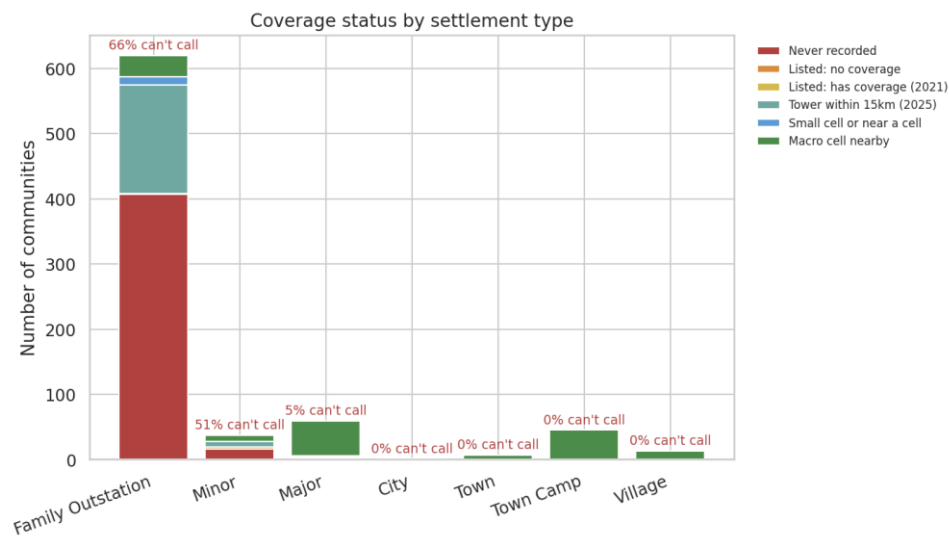


Figure A6. Phone coverage category by settlement type.

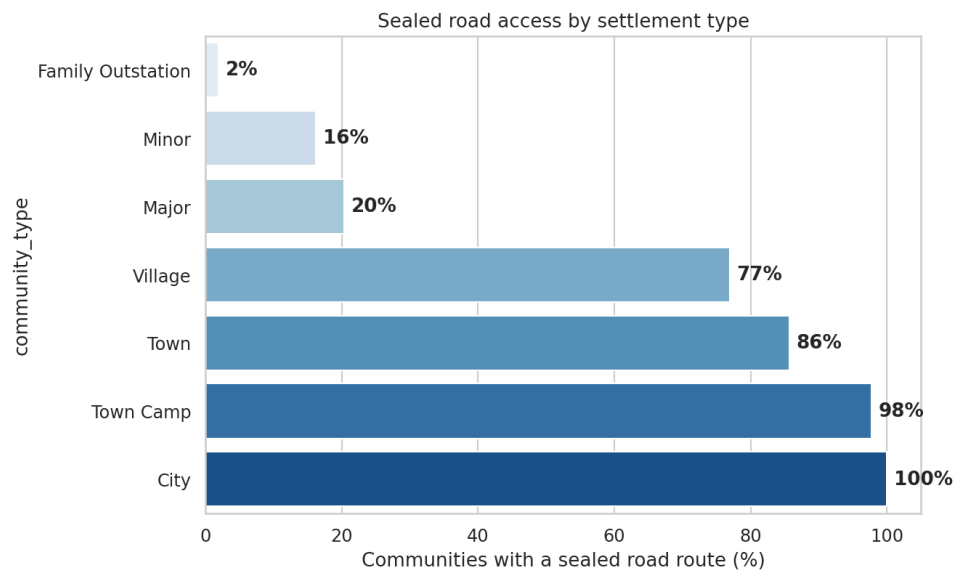


Figure A7. Sealed-road access by settlement type.

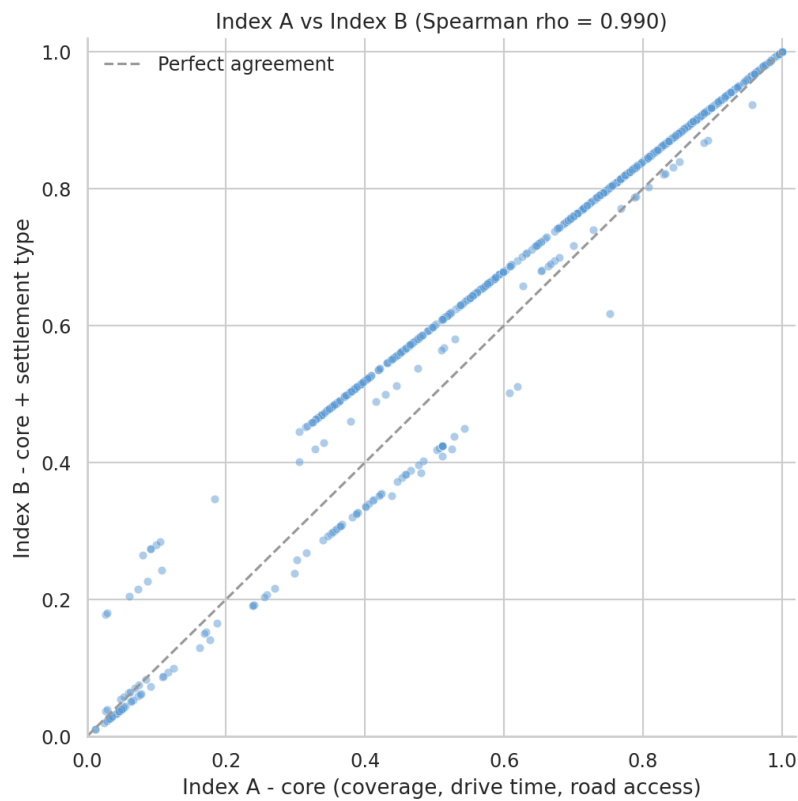


Figure A8. Agreement between Index A (core components) and Index B (core plus settlement type). $\rho = 0.990$.

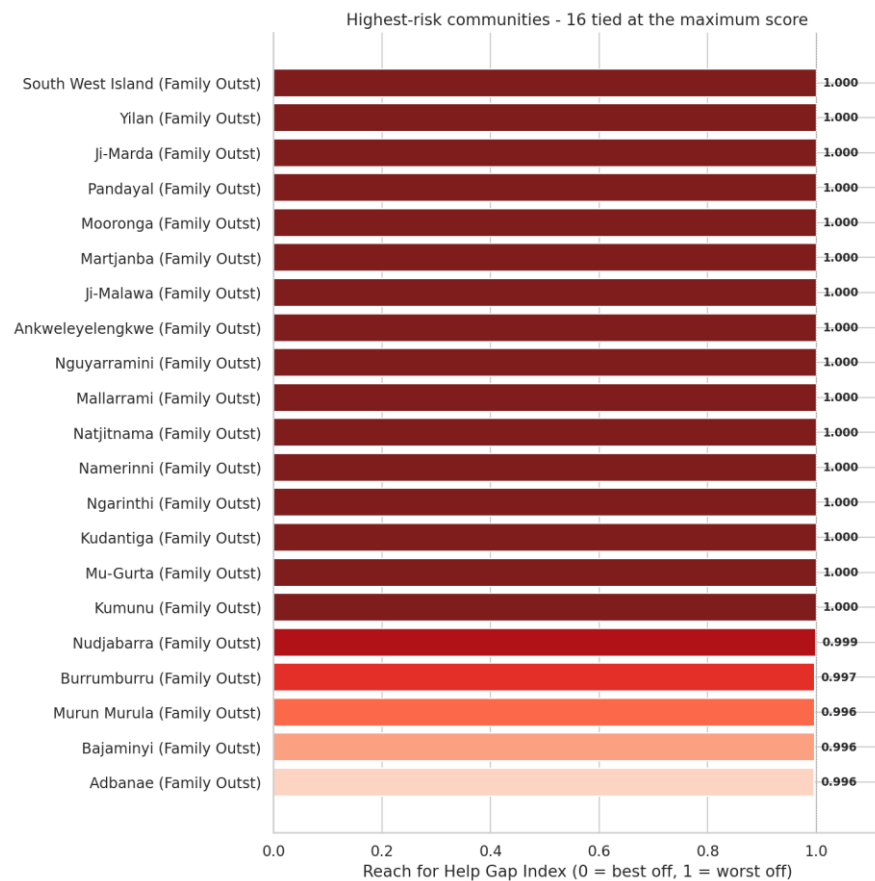


Figure A9. The highest-risk communities by Gap Index; 16 communities are tied at the maximum score of 1.000.

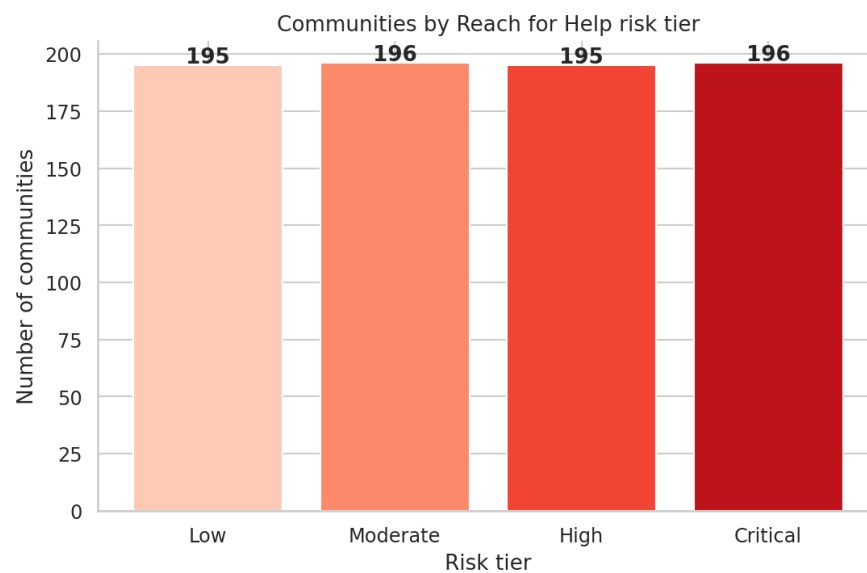


Figure A10. Distribution of NT communities across the four Gap Index risk tiers.